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Chapter 5

EDI Application Interface Overview



Introduction

Electronic Data Interchange is not a new concept. In the early 1960s many large multi-nationals, such as Philips and General Motors, started to transfer internal information and instructions electronically between different application systems, whether by tape or telecommunications.

Since then, technology and business practice have evolved to the point where EDI is virtually a condition of trade in some business sectors.

The application of EDI has accelerated as companies have come to realise that EDI can bring many measurable benefits:

Speed

EDI sends and receives documents quickly. There is none of the delay associated with mailing the documents.

Accuracy

Since EDI transmits documents electronically, there is no need to re-enter the information, thus reducing keying errors.

Cost Savings

The quick turn-around of documents reduces inventory overhead and increases the available cash flow. EDI also reduces mailing costs.

Productivity

Employees are free to perform different tasks. They are no longer needed to enter documents or address, stamp and mail envelopes.

The EDI solution has two closely integrated components. These are:

- The EDI Application Interface module
- Premenos EDI product

Together, these products satisfy the requirements of a successful EDI installation, namely:

- Controlled Application Interfaces
- Standard Message Format Support
- Conversion Facilities
- Communications Management Facilities

EDI will provide support for the most widely-used Message Standards and Value Added Networks. It will also give mailbox access and document conversion facilities. The Application Interface module supplies the facility to import and export documents - in a controlled fashion - to and from the standard Sales Order Processing and Purchase Management modules.

Overview

EDI Application Interface enables the import and export of documents to and from Sales Order Processing and Purchase Management. These documents, traditionally sent and received in a paper form, can now be exchanged as electronic messages. The application will support:

- Inbound sales orders and order amendments
- Outbound acknowledgements
- Outbound advice of despatch
- Outbound sales invoices and credits
- Outbound purchase orders and amendments
- Outbound vendor schedules
- Outbound catalogues
- Inbound acknowledgements
- Inbound advice of despatch
- Inbound supplier invoices (onto the Accounts Payable log)
- Inbound schedule orders
- Inbound advanced shipment notes

The application can be used in two ways.

The application is designed to work very closely with the Premenos EDI product, EDI/400. This will cater for the conversion of messages into and out of standard message formats. The Application Interface module adds new function for users of Sales Order Processing, Vendor Scheduling and Purchase Management in the shape of direct calls to EDI. So, for example, one interface option will allow the preparation of electronic purchase orders, and will also invoke a routine in EDI to convert those orders into standard message formats for onward transmission to suppliers. This close-coupled link to EDI removes the need for repeated manual intervention in the conversion and transmission process, which can be time consuming, and thus costly.

The second potential use for the interface module is as an electronic batch transaction facility. For example, it is common in certain industries for sales orders to be captured by the field sales force on portable microcomputers. Traditionally, these have then been re-entered onto the central Sales Order Processing application. With the arrival of the Application Interface module, this is no longer necessary. Using existing facilities, the orders can be loaded up onto the host system, and taken into the order bank, in a controlled fashion, using the Application Interface module. There are many similar potential uses for the module.

The control aspect must be emphasised. This module applies the same validation logic to inbound and outbound messages as is used in the conventional operation of the system. Full exception and audit reporting is provided.

Premenos EDI

EDI provides the following functions:

- Reformatting of messages between the private format of the Application Interface module and the formats of the major international EDI Standards
- Communication modules to enable direct and straightforward connection to all the principal value added networks
- Support for point-to-point communications where a value added network is not in use
- Full transaction logging, error recovery and security features

Support Of EDI Standards

EDI will support the following national and international EDI Standards:

- ANSI X12
- INCOME
- EDIFACT
- ODETTE
- Tradacoms
- TDB
- US

In addition, EDI contains a standard modification tool, which will allow modification of standards already defined to the product, and definition of new standards.

Value Added Network Support

EDI is supplied with a module enabling connection to the Premenos Information Network. In addition, further communication modules will be available to enable connection to any, or all of the following value added networks:

- INS/TRIDENT
- AT&T Easylink
- GEIS network services
- McDonnell Douglas network services
- Control Data/Redinet
- Sterling/Ordernet Version 3
- Kleinschmidt network services
- TranSettlement/Transend
- General Motors/GMNET
- Ford/SOLMIS

As well as the value added networks (VAN) listed above, EDI will allow direct point-to-point communications between systems, where a VAN is not used.

Relationship To Other Applications

The following applications are pre-requisite to the EDI Application Interface:

- Inventory Management
- Sales Order Processing (for sales documents only)
- Purchase Management (for purchasing documents only)
- Vendor Scheduling (for vendor schedules documents only)
- Accounts Payable (for supplier invoices)
- Advanced Receiving (for advanced shipment note documents only)
- Application Manager

The application is designed to be able to add translated documents into the mailbox send queue in the EDI application automatically. This requires, however, the trading partner being defined to EDI.

Application Configuration

To set up the additional data required for sending and/or receiving documents via the EDI Application Interface. The main elements to be created are:

- Documents
- Trading partners
- Location references
- Translation maps
- Code translation tables

Documents

All document types to be sent via EDI must be defined in the Application Interface, e.g. purchase order, sales invoice. Within this, the document ID must be specified for each EDI message standard used.

These definitions will be shared by all companies defined to the application.

Trading Partners

All trading partners (customers or suppliers) who are to be communicated with via EDI must be defined in the Application Interface.

Within this, the document types to be sent to each partner must be selected from those defined above. At this point it should be specified if the document is to be printed and/or sent via EDI for this trading partner.

Further details must be defined for the trading partner in the EDI application. These are required to support the document translation and communication functions in EDI.

Location References

Where delivery is required to multiple locations, a set of agreed location reference codes must be maintained in order to establish the correct delivery address.

Translation Maps

A translation map is required for each trading partner/document combination, whether they are sent or received. This is used by EDI to map data sent via EDI from the Application Interface database into the appropriate EDI message standard, and vice-versa for data received.

Code Translation Tables

A code translation table is required to convert codes, e.g., units of measure, sales tax, into the equivalent codes recognised by the appropriate message standard when sending documents.

Conversely, they are used to translate EDI standard codes into codes for receiving documents.

For more details on setting up trading partners see Trading Partners Maintenance.

Processing Cycle

The following example is provided to illustrate how the EDI processing cycle is supported by Premenos products. The example given is that of an incoming sales order. The procedure for outgoing messages is similar.

Although each step can be executed separately, the products can be configured to allow steps 1, 2 and 3 to be combined into one option leaving step 4 to be executed separately.

Step 1 Access Value Added Network

Having agreed with the applicable electronic trading partners which VAN is to be used to exchange messages, you can define it to EDI. This definition tells EDI how the connection to the VAN should be established.

EDI supports binary synchronous communications with all the networks for which communications modules are provided. For Premenos' IN, full SNA communications are supported. The selected protocols can be used over leased or dialled lines. In the case of dialled connections, either autodial or manual dial is supported. During set-up, other parameters concerning communications with the network will be recorded. EDI stores these parameters with the telephone numbers for dialled connections.

For examples in this product information it is assumed that EDI has established the connections between the AS/400 and an IN.

Step 2 Transfer Documents

Once the communications session is started, EDI will use the previously defined parameters to control the conversation with the network. EDI will then collect any messages sitting in the network mailbox and transfer them down to its own mailbox on the AS/400. It will also check to see if there is any outbound mail (e.g. acknowledgements) that should be sent via IN, and if there are it will transfer them up to the network mailbox. IN will then distribute these to the appropriate trading partners. EDI maintains an on-line log containing the details of each communications session.

Once the transfer process is complete, EDI will - if appropriate - disconnect from the network.

The received message will now be sitting in the EDI mailbox. It can be displayed, printed or deleted while it is there. It can also be translated and moved into the Application Interface Module.

Step 3 Translate To In house Format And Files

Before EDI can successfully translate an incoming order out of its EDI format and into the in-house format, a number of definitions must have been set up within EDI:

- **Trading Partner Transaction Type**

This defines a unique business relationship regarding the two trading partners concerned in the transaction of a particular document, in this case an order. EDI uses this information when handling received messages. The information is also used when addressing and sending outbound messages.

EDI also uses the transaction type to identify the translation map that should be used to translate the document (whether inbound or outbound). The transaction type records other information regarding the relationship that has been agreed with a trading partner, such as the network over which the document should be exchanged, the trading partner's network ID, and version and release values for the standard used.

- **Application Interface File Definitions**

The Application Interface files that will receive the incoming order must be defined to EDI as 'user files'. During translation, EDI directs the translated data to these files.

- **Translation Map**

The relationship between the incoming Tradacoms order and the Application Interfaces files has to be defined. To do this, a 'map' must be created. This map lies at the heart of the translation process. EDI moves data contained in the Tradacoms standard order into the corresponding Application Interface files. A map is required for all EDI documents that are to be sent or received. This is true of all EDI standards.

- **Code Translation Tables**

EDI standards will often have pre-defined codes for certain data elements. These must be translated so as to be accepted as valid. Similarly, if our trading partner uses different values (e.g. UOM, delivery address or product codes), they must also be translated into values acceptable to the particular application. Hence code translation tables. EDI uses these tables to translate standard and trading partner-specific codes into values that are acceptable.

The translation process takes all these elements into account. When completed, the translated order will be written to the appropriate Application Interface file.

Step 4 Application Interface

Once EDI has successfully translated the incoming order and placed it in the Application Interface file; an interface option can be executed to place it on the Sales Order Processing order bank. This interface program will perform the same validation upon the incoming orders that is applied normally to orders entered to the system via the keyboard. Most exceptions will cause the order to be suspended for further investigation. In the case of major exceptions, the order will be rejected. Full exception and control reports are provided.

The logic described in this example is applied to all the documents (whether inbound or outbound) supported by EDI and the EDI Application Interface. In the case of outbound documents (e.g. purchase orders), most of the validation will already have been carried out by the application generating the document. For outgoing documents, EDI and the Application Interface examine the validity of the document as an EDI message, deciding whether they should be sent via EDI and/or printed.

